Problem Set 7 — Survival analysis

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show derivations. Interpret every number you report in context, with units, and say which assumption each conclusion depends on. Notation as in Chapter 20: $y_i = \min(T_i, C_i)$, $\delta_i = \mathbf{1}\{T_i \le C_i\}$, $d = \sum_i \delta_i$, hazard h, cumulative hazard H, survival S; in the Cox model the score is written $U(\beta)$. All data are in the R package `survival`. Use `set.seed(2027)` for anything random.

```
library(survival)
```

1 Type II censoring and an exact interval

Ten identical components are put on test at time 0 and the test is **stopped at the 6th failure**. The failure times, in hours, are 12, 31, 45, 58, 77, 102; the other four components are still working at 102 h. Assume exponential lifetimes with rate λ.

(a) Write the likelihood of λ, and show that it is proportional to $\lambda^r e^{-\lambda T}$, where r is the number of failures and T the *total time on test* (including the survivors). Find $\hat\lambda$ and $\mathrm{se}(\hat\lambda)$ and compute them.

(b) Let $t_{(1)} < \cdots < t_{(r)}$ be the ordered failure times and $t_{(0)} = 0$. Show that the *normalised spacings* $D_i = (n - i + 1)\{t_{(i)} - t_{(i-1)}\}$, $i = 1, \dots, r$, are iid $\mathrm{Exp}(\lambda)$, that $T = \sum_{i=1}^{r} D_i$, and hence that $2\lambda T \sim \chi^2_{2r}$.

(c) Use (b) to derive an exact 95% confidence interval for λ and compute it. Compare it with the Wald interval on the log scale, $\hat\lambda \exp(\pm 1.96/\sqrt{r})$.

(d) Show that $\mathrm{E}[\hat\lambda] = \lambda\, r/(r-1)$ and give the unbiased estimator. Check by simulation (20 000 replications, $\lambda = 0.01$, $n = 10$, $r = 6$) that the MLE is biased upwards, and report the simulated mean.

(e) The same data could have come from a *Type I* study (stop at a fixed time $c = 102$ h) in which 6 components happened to fail before c. Does the likelihood change? Does the exact interval of (c) remain valid?

2 A Gompertz hazard

Mortality of adults is often modelled with the **Gompertz** hazard $h(t) = a\, e^{bt}$, $a > 0$, $t \ge 0$.

(a) Derive $H(t)$, $S(t)$ and $f(t)$.

(b) Show that for $b > 0$ the distribution is proper ($S(t) \to 0$), that $b = 0$ is the exponential distribution, and that for $b < 0$ the distribution is *defective*: $S(\infty) = \exp(-a/|b|) > 0$ (a "cure fraction"). Evaluate $S(\infty)$ for $a = 0.01$, $b = -0.05$.

(c) Take $a = 0.01$ and $b = 0.05$ per year, t in years of age since 40. Compute (by hand, with a calculator) $S(60)$, the median $t_{0.5}$, and $P(T > 70 \mid T > 60)$.

(d) Show that the Gompertz family is closed under proportional hazards: if $h_0(t) = ae^{bt}$ then $h_0(t)e^{x\beta}$ is Gompertz with parameters $(ae^{x\beta}, b)$. A covariate has HR $= 2$. Compute the median under HR $= 2$ and compare the ratio of the two medians with $1/2$. What does this say about reading "a hazard ratio of 2" as "half the median survival"?

(e) For right-censored data (y_i, δ_i) write the log-likelihood $\ell(a, b)$. For fixed $b \neq 0$ find $\hat a(b)$ and the profile log-likelihood of b.

(f) The data frame `pbc` (Mayo Clinic primary biliary cirrhosis trial) has `time` in days and `status` 0 = censored, 1 = transplant, 2 = death. Use the 312 randomised patients (rows 1–312), time y in **years**, event = death (transplant is treated as censoring). Fit the Gompertz model by maximum likelihood with `optim()` (parametrise by $\log a$ and b), report $\hat{a}$, $\hat{b}$ and the maximised log-likelihood, test $H_0 : b = 0$ with a likelihood-ratio test, and compare AIC with the exponential and the Weibull models (use `survreg()`). Compare $\hat{S}(5)$ and $\hat{S}(10)$ of the Gompertz fit with Kaplan–Meier.

```
pb <- pbc[1:312, ]
y_pb <- pb$time / 365.25
d_pb <- as.integer(pb$status == 2)
c(n = length(y_pb), deaths = sum(d_pb), person_years = sum(y_pb))
```

```
#>            n       deaths person_years
#>          312          125         1714
```

3 Kaplan–Meier and the log-rank test by hand (ovarian cancer)

The data frame `ovarian` has 26 women with advanced ovarian cancer randomised to two treatments (`rx` 1 or 2); `futime` is the follow-up in days and `fustat` the event indicator (1 = death). The 13 patients on treatment 1 have times (+ = censored)

$$59,\ 115,\ 156,\ 268,\ 329,\ 431,\ 448^{+},\ 477^{+},\ 638,\ 803^{+},\ 855^{+},\ 1040^{+},\ 1106^{+}.$$

```
ov <- survival::ovarian
table(ov$rx, ov$fustat)
```

```
#>
#>     0 1
#>   1 6 7
#>   2 8 5
```

(a) Build the Kaplan–Meier table for treatment 1 by hand $(t_j, n_j, d_j, \hat{S}(t_j))$. Compute Greenwood's standard error and a 95% log-log confidence interval for $\hat{S}(638)$, and the median survival.

(b) With R, estimate the Kaplan–Meier curves by treatment, report the medians with 95% intervals, and plot the curves with their log-log confidence bands. Why does treatment 2 not have a median?

(c) For the two groups together, compute by hand n_j, n_{1j}, d_j, E_{1j}, V_{1j} at the first two event times (treatment 1 is group 1). Then compute the whole log-rank statistic with a loop, and check it with `survdiff()`.

(d) Repeat the test with the weights $\hat{S}(t_j-)$ (`rho = 1`). Do the conclusions change?

(e) Fit a Cox model for `rx` and report the hazard ratio with its interval. The data contain 12 deaths. Use Schoenfeld's formula $d = 4(z_{0.975} + z_{0.80})^2/(\log \mathrm{HR})^2$ to find how many deaths are needed to detect the *observed* hazard ratio with 80% power at the 5% level (1:1 allocation), and comment.

4 Cox partial likelihood by hand, ties, and the log-rank test

Six subjects have times 3, 5, 5, 7^{+}, 9, 12 and a binary covariate $x = (0, 1, 0, 1, 1, 0)$. Consider $h(t \mid x) = h_0(t)e^{x\beta}$ and write $z = e^{\beta}$. Two subjects die at $t = 5$ (subjects 2 and 3).

(a) List the risk sets. Write the Breslow partial log-likelihood $\ell_B(\beta)$ and the contribution of the tied pair at $t = 5$ under the Efron and the exact methods (as in Chapter 20, Section "Tied event times").

(b) Using the Breslow version, compute $U(0)$ and $I(0)$, the score statistic $U(0)^2/I(0)$, and a one-step Newton estimate $\beta^{(1)}$ from $\beta = 0$.

(c) Compute the log-rank statistic (group 1 = subjects with $x = 1$) by hand, with the hypergeometric variance. Compare it with the Breslow score statistic in (b), and explain *exactly* where the two differ.

(d) With `coxph()` fit the three tie methods. Which of them reproduces the log-rank statistic as its score test? Explain why, using the fact that, under $H_0 : \beta = 0$, the set of subjects failing at a tied time is uniformly distributed over all subsets of the risk set of that size.

(e) With $\hat{\beta}$ from the Breslow fit, compute the Breslow estimator $\hat{H}_0$ at the event times and predict $S(5 \mid x = 1)$ and $S(5 \mid x = 0)$.

```
toy <- data.frame(y = c(3, 5, 5, 7, 9, 12), d = c(1, 1, 1, 0, 1, 1), x = c(0, 1, 0, 1,
↪  1, 0))
toy
```

```
#>    y d x
#> 1  3 1 0
#> 2  5 1 1
#> 3  5 1 0
#> 4  7 0 1
#> 5  9 1 1
#> 6 12 1 0
```

5 A Cox model for primary biliary cirrhosis

Use the 312 randomised patients of `pbc` (rows 1–312) with time to death in years (transplant is censored), as in Problem 2. Covariates: treatment `trt` (D-penicillamine vs placebo), `age`, `sex`, bilirubin `bili` (mg/dl), `albumin` (g/dl), prothrombin time `protime` (seconds) and `edema` (none, controlled, untreated).

```
pb <- pbc[1:312, ]
pb$yrs <- pb$time / 365.25
pb$death <- as.integer(pb$status == 2)
pb$trt <- factor(pb$trt, 1:2, c("D-penicillamine", "placebo"))
pb$edema <- factor(pb$edema, c(0, 0.5, 1), c("none", "controlled", "untreated"))
```

(a) Report the number of deaths, the median follow-up by the reverse Kaplan–Meier method, the Kaplan–Meier medians by treatment and the log-rank test for treatment. What do you assume when you treat transplants as censored? Is that assumption plausible?

(b) Fit the Cox model `Surv(yrs, death) ~ trt + age + sex + log(bili) + albumin + log(protime) + edema`. Give a table with the hazard ratio and 95% interval for: placebo vs D-penicillamine; 10 additional years of age; doubling of bilirubin; 1 g/dl more albumin; a 10% increase of prothrombin time; untreated edema vs none.

(c) Test the effect of edema (2 df) with a likelihood-ratio test. Is the treatment effect significant? How would you describe, in words, what the data say about treatment?

(d) Check the functional form of bilirubin: compare the AIC with `bili` instead of `log(bili)`, and plot the martingale residuals of a model *without* bilirubin against `bili` and against `log(bili)` with a smoother. Which scale is preferable?

(e) Check proportional hazards with `cox.zph()` and plot the scaled Schoenfeld residual for the covariate with the smallest p-value. Comment, taking into account that eight tests are performed.

(f) Predict the 5-year survival of a 50-year-old woman on placebo with albumin 3.5 g/dl, prothrombin time 10.6 s and no edema, for bilirubin 1 and 6 mg/dl.

(g) Write a paragraph (at most 120 words) for the clinicians reporting the main findings, with hazard ratios and intervals, avoiding causal claims about the prognostic factors.

6 Non-proportional hazards in breast cancer (Rotterdam)

The data frame `rotterdam` has 2982 women with primary breast cancer (Rotterdam tumour bank). Use time to death `dtime` (days; we convert to years) and `death`. Covariates: `age`, tumour `size` (<=20, 20-50, >50 mm), `grade`, number of positive nodes `nodes`, progesterone receptors `pgr` (fmol/l), `hormon` and `chemo` therapy.

```
rt <- survival::rotterdam
rt$yrs <- rt$dtime / 365.25
rt$lnod <- log1p(rt$nodes)       # log(1 + nodes)
rt$lpgr <- log1p(rt$pgr)         # log(1 + progesterone receptors)
```

```r
c(n = nrow(rt), deaths = sum(rt$death), median_follow_up_yrs = median(rt$yrs))
```

```
#>                   n              deaths median_follow_up_yrs
#>            2982.000            1272.000               7.224
```

(a) Fit `Surv(yrs, death) ~ age + size + grade + lnod + lpgr + hormon + chemo` and test proportional hazards with `cox.zph()`. Which covariates violate it? Plot the scaled Schoenfeld residuals of `lpgr` and describe $\beta(t)$.

(b) Split the follow-up at 5 years with `survSplit()` and fit a model in which the coefficient of `lpgr` differs before and after 5 years (use `lpgr:strata(per)`). Report the two hazard ratios per unit of `lpgr` with intervals, and the AIC against the model of (a).

(c) Fit a smooth time-varying effect $\beta(t) = a + b \log t$ with `tt()`. Report the implied hazard ratio per unit of `lpgr` at $t = 1$, 5 and 10 years, and its AIC.

(d) Split at 3 and 6 years (three periods). Does the test of proportional hazards for the new model improve for `lpgr`? Which model do you prefer?

(e) A colleague stratifies by progesterone group (`cut(pgr, c(-1, 0, 100, Inf))`) instead. What does this model estimate, what does it not, and can its AIC be compared with the models in (a)–(d)?

(f) Write two sentences for the report: how does the prognostic information of progesterone receptors change with time since surgery? Offer one possible biological or design explanation, and say what the analysis cannot tell.

7 Survival as Poisson regression: the piecewise exponential model

Let the follow-up be cut into intervals $I_k = (\tau_{k-1}, \tau_k]$, $k = 1, \dots, K$, and assume that the hazard is constant within each interval: $h(t \mid x) = \exp(\alpha_k + x^\top \beta)$ for $t \in I_k$.

(a) For subject i let e_{ik} be the time spent in I_k and $d_{ik} \in \{0, 1\}$ the indicator that the subject died in I_k. Show that the log-likelihood is $\sum_{i,k} \{d_{ik} \log \mu_{ik} - \mu_{ik}\} + \text{const}$ with $\mu_{ik} = e_{ik} \exp(\alpha_k + x_i^\top \beta)$, that is, a Poisson log-likelihood with offset $\log e_{ik}$. What is $\hat\alpha_k$ when there are no covariates?

(b) Use the `colon` data (chemotherapy trial for colon cancer; keep `etype == 2`, death; time in years). Treatment `rx` has three levels (Obs, Lev, Lev+5FU). Use the complete cases of `rx, age, sex, node4, extent` (the data frame `co` below). Split the follow-up at years $1, \dots, 7$ with `survSplit()`, create the exposure `e = yrs - tstart` and fit the Poisson model `status ~ factor(interval) + rx + age + sex + node4 + extent + offset(log(e))`.

(c) Fit the Cox model with the same covariates. Compare the coefficients of the two models in a table. Why are they so close?

(d) Test the constant-hazard (exponential) model against the piecewise exponential with a likelihood-ratio (deviance) test. Describe the shape of the baseline hazard from the interval coefficients.

(e) Fit a Weibull model with `survreg()` and convert its coefficients to the proportional-hazards scale ($-\hat\gamma/\hat\sigma$). Is the Weibull a good description of this baseline hazard?

(f) Interpret the effect of Lev+5FU vs Obs in terms of the death rate per person-year, with a 95% interval.

```r
co <- subset(survival::colon, etype == 2)
co$yrs <- co$time / 365.25
co$node4 <- factor(co$node4, 0:1, c("<=4", ">4"))
co$extent <- factor(co$extent)
co <- na.omit(co[, c("id", "yrs", "status", "rx", "age", "sex", "node4", "extent")])
c(n = nrow(co), deaths = sum(co$status), person_years = sum(co$yrs))
```

```
#>           n      deaths person_years
#>         929         452        4247
```

8 Critique and communicate

An abstract contains the following statements about a randomised trial in 120 patients with a rare cancer (60 per arm):

> *"Kaplan–Meier median survival was 18.2 months on Drug A and 12.1 months on placebo (log-rank $p = 0.047$). The Cox hazard ratio was 0.68 (95% CI 0.46 to 1.01). Fourteen patients withdrew from the study because of adverse events (all in the placebo arm) and were excluded from the analysis. Proportional hazards was verified because the log-rank test was not significant for the interaction with sex. Drug A reduces the risk of death by 32% and extends life by 6.1 months. The mean survival time, estimated from the Kaplan–Meier curve, was 21.4 months (placebo 14.0 months). We conclude that Drug A significantly improves survival."*

(a) Identify at least six problems of statistics or interpretation in this abstract. For each, say why it is a problem and what would be correct.

(b) Rewrite the results in at most 120 words, as they should be reported.

(c) Write a summary of at most 150 words for a patient association (no jargon: no `hazard'',`confidence interval'' or "censored''). It must be honest about uncertainty.

(d) List three further analyses you would ask for before accepting the conclusion.